

Istiaq Bin Mahmud

CONTACT INFORMATION	Chittagong, Bangladesh Phone: (+880) 1537256051	istiakmahmod842@gmail.com istiakbinmahmod.com
EDUCATION	Bangladesh University of Engineering and Technology B.Sc in Computer Science and Engineering - Thesis Supervisor: Dr. Muhammad Masroor Ali - Coursework: <i>Discrete Mathematics, Data Structures and Algorithms, Digital Logic Design, Theory of Computation, Database, Computer Architecture, Compiler, Microprocessors, Microcontrollers, and Embedded Systems, Operating System, Artificial Intelligence, Computer Networks, Computer Security, Computer Graphics, Basic Graph Theory, Machine Learning</i> etc.	CGPA : 3.63/4.00 April 2018 - May 2023
STANDARDIZED TEST SCORES	GRE <i>Quant: 165, Verbal: 155, AWA: 3.5</i>	320
	TOEFL <i>Reading: 26, Listening: 27, Writing: 25, Speaking: 23</i>	101
PROFESSIONAL EXPERIENCE	Optimizely , Dhaka, Bangladesh <i>Software Engineer II</i> - Led the development and architectural refinement of the Edge Delivery platform, significantly improving system throughput via advanced query optimization and distributed caching strategies. - Engineered and deployed algorithmically driven core features, notably Contextual Multi-Armed Bandits for experimentation and a dynamic Visual Editor. - Architected the modernization of legacy microservices, successfully migrating large-scale infrastructure from Python 2/Webapp2 to Python 3/Flask. - Implemented robust Datadog observability pipelines across distributed microservices, managing full-stack deployment, DevOps practices, and system reliability. <i>Tools & Technologies:</i> Python (Flask), TypeScript, React, GCP, Cloudflare Workers, Akamai EdgeWorkers, Datadog, Docker, CI/CD, Git.	<i>October 2023 – Present</i>
	Computer Network Systems Limited , Dhaka, Bangladesh <i>Assistant Software Engineer</i> - Contributed to the large-scale modernization of a legacy Bangladesh government portal, refactoring architecture into a scalable Spring Boot and React application. <i>Tools & Technologies:</i> Java (Spring Boot), React JS, TypeScript, Oracle DBMS.	<i>June 2023 – August 2023</i>
HONORS AND AWARDS	Dean's List Award General Merit Scholarship, Higher Secondary School Certificate Examination Talentpool Merit Scholarship, Secondary School Certificate Examination 3rd position, Bangladesh Chemistry Olympiad Division, Divisional round (Chittagong, Bangladesh)	2018, 2019 2017 2015 2015
RESEARCH EXPERIENCE	What Kind of Contracts Do LLM APIs Need? – Design By Contract for LLM APIs Developed a novel contract verification framework tailored for Large Language Model (LLM) APIs to systematically prevent integration bugs.	

- Conducted an empirical analysis of 200+ API contract violation cases, extracting and evaluating 40+ LLM-specific instances to build a comprehensive taxonomy.
- Engineered an automated design-by-contract tool for the OpenAI ecosystem, extending prior *DLContracts* methodologies to enable pre-deployment violation detection and improve LLM application reliability.

Undergraduate Thesis [\[PDF\]](#)

Supervisor: Dr. Muhammad Masroor Ali | Collaborator: Soham Khisa

- Proposed and developed an ontology-based newspaper categorization system to extract structured metadata from unstructured HTML news articles.
- Utilized the IPTC Media Topics taxonomy (comprising over 1,200 hierarchical terms) to map semantic relationships and accurately categorize text corpora based on complex ontology concepts.

ACADEMIC PROJECTS

Moodle LMS [\[Code\]](#) [\[Demonstration\]](#)

We developed a Moodle LMS (Learning Management System) for our software engineering project. We utilized the MERN stack (MongoDB, Express, React, Node) to develop a web-based LMS platform. This website was inspired by moodle.cse.buet.ac.bd.

Blood Donating Platform [\[Code\]](#)

We designed and developed a blood donation app as an undergraduate project. The website allows users to sign up as donors, specify their blood type, and provide their contact information. When a blood drive or a hospital requests blood, the app sends notifications to eligible donors in the area, allowing them to quickly and easily respond to the call for donation.

Retro Shooting Game [\[Code\]](#) [\[Demonstration\]](#)

We built a retro shooting game using C++ and the ATMEGA32 microcontroller. This game features 2D graphics, simple and fast-paced action, and a focus on high scores and player skill. The goal is usually to achieve the highest score possible, either by surviving for as long as possible or by collecting points for defeating enemies.

TECHNICAL SKILLS

- Languages: Python, Java, JavaScript, C++, HTML5, CSS, Bash
- Frameworks & Libraries: Flask, React, Node.js, Django, Spring Boot
- Databases: GCP Datastore, Spanner, Oracle, MySQL, MongoDB
- Cloud & Infrastructure: GCP, Cloudflare Workers, Akamai EdgeWorkers, Docker
- Tools & Platforms: Git, GitHub, Datadog, CI/CD, L^AT_EX, NS3, PacketTracer, Wireshark